CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A sonar signal gives echoes from two surfaces at 0.8 s and 1.0 s. The separation between the surfaces (v = 1500 m/s) is: 1

(a) 600 m **(b)** 75 m **(c)** 150 m **(d)** 300 m

2. A car goes from rest with acceleration a for time t, then retards uniformly with 2a till it stops. The total time is: 1

(a) t **(b)** 3 t **(c)** 1.5 t **(d)** 2 t

3. Assertion (A): Action and reaction forces never cancel each other. Reason (R): They act on different bodies. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): The kinetic energy of a body becomes four times when its speed is doubled. Reason (R): KE is proportional to the square of the speed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): The speed of sound is greater in solids than in gases. Reason (R): The particles in solids are closely packed and strongly bound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A man walks to a market at 4 km/h and returns at 6 km/h. The average speed is: 1

(a) 5 km/h **(b)** 5.5 km/h **(c)** 4.5 km/h **(d)** 4.8 km/h

7. A force of 12 N acts on a body of mass 3 kg for 5 s starting from rest. The final speed is: 1

(a) 12 m/s **(b)** 4 m/s **(c)** 20 m/s **(d)** 60 m/s

8. A body falls from rest from a height h. At what height is its speed half of the final speed? 1

(a) h/8 **(b)** 3h/4 **(c)** h/2 **(d)** h/4

9. A person standing at 340 m from a cliff fires a gun. The echo is heard after: 1

(a) 4 s **(b)** 2 s **(c)** 0.5 s **(d)** 1 s

10. Assertion (A): The average velocity of a body in a round trip is zero. Reason (R): The net displacement is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. A heavy box is pushed along the floor with a force F. Which of the following would make the push easier? 1

(a) reducing the friction by rollers **(b)** using a rough floor
(c) increasing the weight **(d)** pushing downward

12. A bob of a simple pendulum is displaced and released from 0.8 m height above the lowest point. The speed at the lowest point is: 1

(a) 2 m/s **(b)** 4 m/s **(c)** 16 m/s **(d)** 8 m/s

13. A sound of 20 degrees Celsius air moves at 344 m/s. The distance travelled in 0.25 s is: 1

(a) 1376 m **(b)** 86 m **(c)** 43 m **(d)** 172 m

14. A stone dropped from a height h takes time t. If dropped from 4h, the time taken will be: 1

(a) t **(b)** 2t **(c)** t/2 **(d)** 4t

15. A ball of mass 100 g is thrown with a speed of 20 m/s horizontally and hits the wall to stop in 0.01 s. The force is: 1

(a) 2 N **(b)** 20 N **(c)** 200 N **(d)** 2000 N

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Explain why a person at the surface of Earth does not move towards the Earth with acceleration g but is at rest. 2

17. An engine develops 2 x 10^5 W power and moves a train at 20 m/s. Find the force of the engine. 2

18. Why are two sounds of the same loudness and pitch from a violin and a sitar distinguishable? 2

19. A bus moving at 15 m/s sees a red light 60 m ahead. Find the minimum retardation needed to stop it just before the light. 2

20. Why does the tension in a rope used to pull a block at constant velocity equal the friction on the block? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A siren emits a sound of frequency 700 Hz. Find the time period, the wavelength in air (340 m/s), and the number of waves reaching a man in 5 s. 3

22. A lift moves upward with acceleration 2 m/s squared for 4 s, then uniformly for 6 s and decelerates at 4 m/s squared. Find the height travelled. 3

23. A hammer of mass 1 kg moving at 10 m/s strikes a nail and stops in 0.01 s. Find the force and the force on the nail. 3

24. An object of mass 2 kg is lifted vertically from rest by a force of 30 N through 3 m. Find the work done by the force, by gravity and the final speed (g = 10). 3

OR

Explain why the sound of a bell is heard for a long time in a stone temple but only for a short time in an open field. 3

25. A car A at 90 km/h overtakes car B moving at 72 km/h. If A is 50 m behind B, find the time taken to be 50 m ahead of B. 3

OR

A force of 30 N acts on a 6 kg body initially at rest for 4 s. Find the final velocity, the change in momentum and the distance covered. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the motion of a ball moving in a vertical circle of radius 1 m with speed 4 m/s, the period, the speed vs velocity, and the average velocity over half a period (pi = 3.14). 5

OR

A cricket ball of mass 160 g approaches the bat at 30 m/s and leaves at 40 m/s in the opposite direction in 0.02 s. Find the impulse and the force, and explain how the force changes if the contact time is doubled. 5

27. A power station delivers 3 MW. Find the energy in kWh in one day and the cost at 5 rupees per unit. Also find the number of 100 W bulbs it can light. 5

OR

Explain why echoes cannot be heard in a small room, but multiple echoes are heard in a big hall, and design the dimensions needed for a hall so that reverberation time is reduced. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND